

OBSTETRICS

Routine 36-week scan: optimizing delivery timing of large for gestational age fetuses



Antonio Farina, MD; Paolo Ivo Cavoretto, MD; Argyro Syngelaki, PhD; Ioannis Mitrogiannis, MD; Ranjit Akolekar, MD; Kypros Herodotos Nicolaides, MD

BACKGROUND: Large for gestational age fetuses present an increased risk of labor and delivery complications, potentially preventable through timely childbirth.

OBJECTIVE: To evaluate the risk for cesarean due to fetal compromise and/or failure to progress in pregnancies with large for gestational age detected at routine 36-week ultrasound scan and to identify optimal birth timing.

STUDY DESIGN: Analysis of prospectively collected data from 2 UK Fetal Medicine centers including ultrasonographic estimated fetal weight at 35 to 36 weeks' gestation and outcomes. Of 107,875 pregnancies, 84,397 were analyzed after excluding prelabor cesareans and estimated fetal weight <10th percentile. The final cohort included 7695 large for gestational age (estimated fetal weight >90th percentile), of which 3384 were inductions and 4311 were spontaneous labors. A competing risk model examined labor type (spontaneous vs induced) and cesarean for fetal compromise and/or failure to progress, assessing cumulative incidence and instantaneous hazard curves.

RESULTS: This study showed that: first, the hazard of cesarean delivery due to fetal compromise and/or failure to progress increases progressively with advancing gestational age with steeper rise after 40 weeks; second, at a given gestational age, the hazard is higher in nulliparous vs parous and

in induction vs spontaneous labor; third, there is a gradient of hazard progression when estimated fetal weight increases from the 90th to above the 95th percentile; fourth, a policy of induction at 38 to 39 weeks for fetuses with estimated fetal weight >95th or 90th to 95th percentile approximates the hazard and cumulative incidence of large for gestational age fetuses delivering spontaneously at 40 weeks; and fifth, induction beyond 41 weeks is associated with a substantially increased risk.

CONCLUSION: The findings support induction of labor at 38 weeks in cases of estimated fetal weight above the 95th percentile and at 39 weeks for estimated fetal weight between the 90th and 95th percentile, as this strategy achieves a risk of cesarean delivery due to fetal compromise and/or failure to progress comparable to that of spontaneous labor at 40 weeks, while avoiding the progressive increase in risk observed beyond 40 weeks. Alternatively, conservative management until 41 weeks may be considered, followed by elective cesarean delivery if spontaneous labor has not occurred by then, to avoid potentially harmful effects of induction at that stage, associated with a marked increase in cesarean risk.

Key words: 36 weeks, cesarean delivery, competitive risks model, EFW, instantaneous smoothed hazard, labor induction, LGA

Introduction

Background

Large for gestational age (LGA), generally defined as estimated fetal weight (EFW) or birthweight (BW) above the 90th percentile for gestational age (GA),¹ is associated with an increased risk of intrapartum maternal-fetal trauma, perinatal morbidity, and both short-term and long-term complications.^{2–4}

A multicenter trial in France reported that in pregnancies with suspected LGA fetuses (>95th percentile), induction of labor (IOL) at 37 or 38 weeks' gestation vs standard care was associated with 70% reduction in the risk of shoulder dystocia and related morbidity, without increasing

the risk of cesarean delivery.⁵ In contrast, a multicenter trial in the United Kingdom reported no difference in the incidence of shoulder dystocia between suspected LGA (>90 customized percentile) undergoing IOL at 38 weeks vs standard care, in the intention-to-treat analysis; however, in the per-protocol analysis, IOL at 38 weeks showed a 40% lower incidence of shoulder dystocia.⁶

The adverse outcome selected as the endpoint in these trials was shoulder dystocia, a rare event that is often challenging to define and diagnose consistently. Evidence also shows that LGA fetuses are more prone to other adverse intrapartum outcomes, such as labor dystocia, prolonged first and second stages, and perinatal hypoxia and asphyxia, which contribute to higher rates of perinatal death and increased incidence of cesarean delivery, making their investigation and prevention highly relevant for improving perinatal outcomes.^{2–4,7}

Objective

We hypothesized that timely induction in LGA fetuses may lower the risk of cesarean delivery for fetal compromise and/or failure to progress. We aimed to quantify the cumulative incidence and hazard of cesarean delivery for these reasons, in induced vs spontaneous labor, analyzing risk profiles by EFW and parity to guide personalized timing of induction.

Methods

Study design, setting, and participants

This was a secondary analysis of prospectively collected data from the Fetal Medicine Research Centre (London, UK) and Medway Maritime Hospital (Kent, UK) in which all women with singleton pregnancies had a routine ultrasound examination at between 35⁺⁰ and 36⁺⁶ weeks' gestation. The recruitment period was from February 2014 to

0002-9378/\$36.00

© 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.
<https://doi.org/10.1016/j.ajog.2026.03.023>



Click Supplemental Materials and Video under article title in Contents at ajog.org

AJOG at a Glance

Why was this study conducted?

In pregnancies with large for gestational age (LGA) fetuses, there is contradictory evidence whether induction of labor before term reduces the risk of birth trauma, such as shoulder dystocia. However, shoulder dystocia is a relatively rare event, and previous studies have overlooked more frequent adverse events, such as intrapartum fetal compromise and labor dystocia. There is therefore a need to reevaluate the optimal timing of birth of LGA fetuses.

Key findings

This study addresses the issue by comparing induced vs spontaneous labor, to determine strategies that effectively reduce cesarean deliveries related to major labor complications. Optimizing the timing of delivery in LGA fetuses through labor induction at 38 to 39 weeks' gestation, guided by parity and ultrasonographic estimated fetal weight (EFW) at 36 weeks, reduces the risk of cesarean delivery while balancing the risks of intrapartum fetal compromise and labor dystocia against fetal maturity. Induction at 38 weeks for EFW >95 percentile and at 39 weeks for EFW 90th to 95th percentile shows greatest benefit.

What does this add to what is known?

The findings support induction of labor at 38 weeks' gestation in cases of EFW above the 95th percentile and 39 weeks for EFW between the 90th and 95th percentile. Alternatively, conservative management until 41 weeks may be considered, followed by discussion of elective cesarean delivery if spontaneous labor has not occurred by then, to avoid potentially harmful effects of induction of labor at that stage.

December 2023. The examination included measurements of fetal head and abdominal circumference and femur length to calculate the EFW by the Hadlock formula.⁸ Ultrasound assessment was performed according to the Fetal Medicine Foundation (FMF) guidelines by expert operators with certificates of competence granted by the FMF undergoing regular performance, audits, and follow-up assessments. This included standard fetal biometry, fetal Doppler studies, and anatomical survey to exclude major fetal abnormalities. Inclusion criteria were singleton pregnancies with spontaneous labor or IOL resulting in live births without major fetal abnormalities. Exclusion criteria were EFW <10th percentile and elective cesarean deliveries.

The analysis primarily compared outcomes between spontaneous labor and IOL, focusing on the cumulative incidence and hazard of cesarean delivery due to fetal compromise and/or failure to progress, in those with EFW

90th to 95th percentile and those with EFW >95th percentile, using the appropriate for gestational age (AGA) group with EFW between the 10th and 90th percentile as the reference. This observational study was reported in agreement with the STrengthening the Reporting of OBservational studies in Epidemiology (STROBE) guidelines.⁹

Variables and outcomes

GA was determined in the first trimester through ultrasound measurement of fetal crown-rump length.¹⁰ The independent variables in this study included maternal characteristics such as parity, body mass index (BMI), height (categorized in 3 groups using a clinically established threshold to preserve interpretability), conception method (natural vs in-vitro fertilization), comorbidities (type I and II diabetes, chronic hypertension, gestational diabetes mellitus [GDM], preeclampsia [PE], and gestational hypertension), and EFW. Classification of cases used

nonsex-specific FMF fetal and neonatal percentile charts that do not adjust for maternal parity; this factor was therefore modeled independently.¹¹ The mode of labor onset, either induced or spontaneous, was also recorded.

Spontaneous labor was characterized by either prelabor rupture of membranes with subsequent uterine contractions or by substantial progressive cervical changes, along with regular uterine contractions or cervical dilation >6 cm.¹² Failure to progress was defined according to American College of Obstetricians and Gynecologists (ACOG)/Society for Maternal Fetal Medicine (SMFM) guidelines. First stage: cervical dilation of ≥ 6 cm with ruptured membranes, accompanied by either ≥ 4 hours of adequate uterine contractions without cervical change or ≥ 6 hours of inadequate contractions with no cervical progression with oxytocin administration; Second stage: 2 hours of pushing in parous and 3 hours in nulliparous.¹³

Indication for IOL was based on individual clinicians' recommendations aligned with local protocols and National Institute for Health and Care Excellence (NICE) guidelines.¹⁴ The procedure for IOL followed local protocols and included the use of prostaglandins, oxytocin, amniotomy, and cervical ripening balloons, in line with established recommendations.^{14,15} Fetal compromise was defined as reported by our group^{16–18} and assessed by the attending physicians based on individual clinical findings, primarily involving abnormalities in fetal heart rate patterns observed via continuous cardiotocography (ie, bradycardia, recurrent-prolonged decelerations, and diminished beat-to-beat variability) with or without meconium-stained amniotic fluid.

The primary outcome was cesarean delivery due to fetal compromise and/or failure to progress, expressed with the cumulative incidence of cesarean delivery in spontaneous labor vs IOL, and smoothed instantaneous hazard (SIH).

Additional indications for cesarean delivery during labor were censors of the models encompassing: antepartum hemorrhage, breech presentation,

maternal request, relevant maternal conditions, placenta praevia, history of cesarean delivery or myomectomy, transverse lie, and uterine rupture. Stillbirths were included, being reported in [Supplemental Figure 1](#) and were censored in the survival models. Neonatal weight was recorded and compared to estimate fetal weight to establish concordance.

Statistical methods

Data were presented by labor type with descriptive statistics. The SIH method was used to evaluate the hazard of fetal compromise and/or failure to progress and its acceleration from 35 to 43 weeks' gestation in LGA fetuses >90th centile with spontaneous labor or IOL and in a group of low-risk AGA (natural conception; no major risk factors or maternal/obstetric complications; normal BMI: 18.5–25 kg/m²). The Fine and Gray competing risks model was used to assess the association between EFW percentile categories (10th–90th, 90th–95th, and >95th), modes of labor onset (spontaneous vs IOL), and the risk of cesarean delivery for fetal compromise and/or failure to progress, with vaginal delivery considered a competing event and cesarean delivery for other indications a censoring event. Expectant management at any given GA includes the possibility of subsequent spontaneous labor or continuation of pregnancy with delivery by induction at a later GA. Proportionality of subdistribution hazards was assessed graphically using cumulative incidence functions (CIFs) and smoothed hazard curves; no clinically relevant violations were observed. The CIF was applied to estimate the incidence of cesarean delivery over time, accounting for competing risks.

Differences in CIF of cesarean delivery for fetal compromise and/or failure to progress were calculated at each gestational week comparing spontaneous labor vs IOL, stratified by parity (nulliparous vs parous) and EFW percentiles. The results were shown with forest plots (percentage points difference; negative values favoring IOL and positive values favoring spontaneous

labor). Percentages of concordance were calculated between EFW and BW centile categories.

Statistical analyses were conducted using Stata (version 16; StataCorp 4905 Lakeway Drive, College Station, Texas 77845, 2019). Significance set at a *P* value <.05 (α error <0.05) with a power of 80%, corresponding to a β error of 0.20.

Role of the funding source

The funding source supported data collection and was not involved in study design, analysis, interpretation of data, in the writing of the report, and in the decision to submit.

Results

Participants, descriptive, and outcome data

A total of 107,875 women were screened. Pregnancies with EFW <10th percentile were excluded, leaving 84,397 pregnancies eligible for analysis. From this cohort, a subset of 7695 cases with an EFW above the 90th percentile was identified (3772 cases with EFW between the 90th–95th and 3923 cases with EFW >95th). Among cases with EFW >90th percentile, 4311 (56%) experienced spontaneous labor and 3384 (44%) underwent IOL. The remaining 76,702 pregnancies were classified as carrying fetuses that were AGA with EFW between the 10th and 90th percentile. [Supplemental Figure 1](#) shows the flowchart for patient screening and selection. All stillbirths in the LGA group were antenatal, apart from one intrapartum case associated with cesarean delivery for fetal compromise.

The descriptive analysis reported refers only to 7695 patients bearing fetuses with EFW >90th percentile. [Table 1](#) summarizes patient characteristics according to the mode of labor. Patients undergoing IOL had a higher BMI and higher rates of nulliparity, in vitro fertilization conception, diabetes mellitus, chronic hypertension, and previous pregnancies with LGA or PE.

[Table 2](#) includes pregnancy outcomes of the 7695 participants, according to the onset of labor. Patients

undergoing IOL had a lower GA at birth and higher time interval to birth, BW and BW percentile, and rates of GDM, PE, and gestational hypertension, which correlate with a higher incidence of cesarean delivery due to fetal compromise and dystocia.

Main results

The overall cumulative incidence of cesarean delivery due to fetal compromise and/or failure to progress in the whole population of 84,397 pregnancies was 16.9%, in AGA vs LGA fetuses it was 15.9% vs 27.3% (*P*<.001), in AGA vs LGA with spontaneous labor was 11.7% vs 19.1% (*P*<.001), and in AGA vs LGA with IOL was 29.5% vs 37.7% (*P*<.001).

The overall IOL rate was 25.3%, and it was 23.5% in AGA and 44.0% in LGA (*P*<.001). [Table 2](#) shows the breakdown of cumulative incidence at the end of follow-up across subgroups defined by EFW and parity. The censor cases were 2308.

In pregnancies with LGA fetuses, the distribution of GA at birth was similar between spontaneous labor and IOL (in both groups, approximately 10% of pregnancies delivering >41⁺⁰ weeks and 50% between 39⁺⁰ and 41⁺⁰ weeks).

Rates of IOL vs spontaneous labor across the group with LGA are shown in [Table 3](#). A series of specific patterns composed of the most frequent combinations of common 7 risk factors are shown (rare patterns not shown). The risk factor combinations analyzed in this table include BMI >30 kg/m², pregnancy complications (ie, GDM, PE, and gestational hypertension), preexisting conditions (ie, diabetes and chronic hypertension), and EFW percentiles (90th–95th or >95th). The subanalysis identifies the factors that substantially influence the likelihood of IOL. As shown, the 3 most frequent patterns, accounting for about 50% of cases, involve BMI >30 kg/m² and/or EFW >90th percentile. The most common pattern, >30 kg/m² combined with EFW >95th percentile, is equally distributed between spontaneous labor and IOL (22.1% vs 23.9%, *P*=.065). In contrast, spontaneous labor is more frequent in cases with EFW 90th to 95th percentile, regardless of BMI, and in

TABLE 1

Summary of characteristics of the 7695 women included in the study according to mode of labor

Variables	Spontaneous labor (n=4311)	Induction of labor (n=3384)	P value ^a
Maternal age (y)	32.1 (5.2)	32.0 (5.6)	.550
BMI (kg/m ²)	31.9 (5.9)	33.9 (6.6)	<.001
Maternal height (cm)	167 (6.5)	167 (6.6)	.760
Ethnic group			.538
Black	12.8	13.3	
East Asian	1.0	0.8	
More than one	2.4	2.0	
South Asian	2.9	2.6	
White	80.8	81.3	
Conception method			<.001
In vitro fertilization	4.9	6.6	
Ovulation drugs	0.8	0.5	
Natural	94.5	93.0	
Cigarette smoking	4.4	4.3	.893
Diabetes mellitus			<.001
Type 1	1.0	1.9	
Type 2	1.1	1.8	
SLE/APS	0.3	0.3	.639
Chronic hypertension	0.8	2.1	<.001
Nulliparous	43.7	47.2	<.001
Previous LGA	12.5	13.1	.005
Previous PE	2.3	3.8	<.001

Data are expressed as mean (SD) or percentages. Continuous variables are expressed as means (standard deviation); categorical variables are expressed as percentages.

APS, antiphospholipid syndrome; BMI, body mass index; LGA, large for gestational age; PE, preeclampsia; SLE, systemic lupus erythematosus.

^a χ^2 or Student *t* test.

cases with EFW >95th percentile without elevated BMI. Conversely, when multiple risk factors are present, such as BMI >30 kg/m², GDM, PE, and pregnancy-induced hypertension in various combinations, the rate of IOL rises considerably.

Figure 1 illustrates the SIH for nulliparous (A) and parous (B) in relation to spontaneous labor vs IOL, in LGA fetuses. Additionally, it presents the SIH for spontaneous labor AGA fetuses in low-risk pregnancies, based on 13,505 nulliparous and 10,492 parous low-risk cases. As shown, irrespective of

parity, SIH increased sharply after approximately 40 weeks in both spontaneous labor and IOL. IOL consistently carried a higher risk, with the risk of spontaneous labor and IOL nearly overlapping in nulliparous women at about 41 weeks, before rising again in IOL. Additionally, it is shown how the risk for LGA fetuses is substantially higher compared to the reference group of low-risk AGA babies.

Table 2 shows the CIF of cesarean delivery due to fetal compromise and/or failure to progress at term (40 weeks) and the end of the follow-up (43 weeks)

according to parity and EFW category. Since the CIF is consistently higher with IOL than with spontaneous labor, establishing IOL at around 39 weeks' gestation for EFW between 90th and 95th percentile aligns the CIF with that observed at 40 weeks in spontaneous labor (approximately 15% in nulliparous and 5% in parous). Matching the CIF similarly for EFW >95th percentile requires IOL 1 week earlier: at 38 weeks, particularly when additional risk factors, such as PE or diabetes, are present.

Figures 2 and 3 show the percentage points differences in the cumulative incidence of cesarean delivery for fetal compromise and/or failure to progress between IOL and spontaneous labor at various GAs, stratified by EFW categories. Negative differences favor IOL; positive differences favor spontaneous labor.

Induction of labor at 38 weeks (estimated fetal weight >95th percentile) and 39 weeks (estimated fetal weight 90th–95th percentile)

Compared to spontaneous labor occurring at $\geq 42^{+0}$ weeks, IOL at 38 or 39 weeks was associated with a theoretical reduction in cesarean delivery for fetal compromise and/or failure to progress of 15.6 percentage points (Figure 2, A) and 18.4 percentage points (Figure 3, A) among nulliparous women with an EFW between the 90th and 95th percentile and above the 95th percentile, respectively. Among parous women, the corresponding reductions were 4.9 and 7.6 percentage points (Figure 3, A and B).

Induction of labor at 39 weeks (estimated fetal weight >95th percentile) and 40 weeks (estimated fetal weight 90th–95th percentile)

For fetuses with EFW >95th percentile, 39 weeks appears to be the latest GA at which IOL may still provide a meaningful benefit but only compared to spontaneous labor theoretically occurring at or after 41⁺⁰ weeks. At 40 weeks, IOL offers minimal benefit, and only for fetuses in the 90th to 95th percentile

TABLE 2

Pregnancy outcomes of the 7695 patients included in the study according to onset of labor (spontaneous vs induced)

Variables	Spontaneous labor (n=4311)	Induction of labor (n=3384)	P value ^a
Gestational age at birth (wks)	39.6 (1.3)	39.4 (1.2)	<.001
Interval (36 wks to birth)	3.6 (2.9)	3.9 (3.1)	.003
Birthweight (g)	3875 (404)	3895 (391)	.030
Birthweight (centile)	82 (17)	83 (16)	<.001
Gestational diabetes	9.0	25.8	<.001
Preeclampsia	1.5	5.5	<.001
Gestational hypertension	1.7	4.9	<.001
Cumulative incidence of cesarean delivery due to fetal compromise and/or failure to progress at term (40 ⁺⁰)			
Nulliparae with EFW 90–95 centile	10.6	22.4	<.001
Pluriparae with EFW 90–95 centile	3.6	8.0	<.001
Nulliparae with EFW >95 centile	13.3	27.5	<.001
Pluriparae with EFW >95 centile	4.6	10.0	<.001
Cumulative incidence of cesarean delivery due to fetal compromise and/or failure to progress at the end of follow-up (43 ⁺⁵)			
Nulliparae with EFW 90–95 centile	28.1	52.6	<.001
Pluriparae with EFW 90–95 centile	10.2	21.7	<.001
Nulliparae with EFW >95 centile	34.2	61.2	<.001
Pluriparae with EFW >95 centile	12.8	26.7	<.001

Continuous variables are expressed as means (standard deviation); categorical variables are expressed as percentages.

EFW, estimated fetal weight.

^a χ^2 or Student *t* test.

TABLE 3

Rates of spontaneous labor vs induction of labor across a series of specific patterns composed of the most frequent combinations of common 7 risk factors

Pattern	BMI >30 kg/m ²	GDM	PE/GH	DM	CH	EFW centile	Spontaneous labor	Induced labor	P value ^a
1	yes	no	no	no	no	>95	954 (22.1%)	809 (23.9%)	.065
2	yes	no	no	no	no	90–95	1107 (25.7%)	575 (17.0%)	<.001
3	no	no	no	no	no	>95	662 (15.4%)	411 (12.1%)	<.001
4	yes	yes	no	no	no	>95	147 (3.4%)	383 (11.3%)	<.001
5	no	no	no	no	no	90–95	1035 (24.0%)	340 (10.0%)	<.001
6	yes	yes	no	no	no	90–95	77 (1.8%)	200 (5.9%)	<.001
7	yes	no	PE	no	no	>95	94 (2.2%)	154 (4.6%)	<.001
8	no	yes	no	no	no	90–95	72 (1.7%)	138 (4.1%)	<.001
9	Yes	no	GH	no	no	>95	24 (0.6%)	60 (1.8%)	<.001
10	Yes	no	GH	no	no	90–95	29 (0.7%)	44 (1.3%)	.0116
11	Yes	no	PE	yes	no	>95	6 (0.1%)	19 (0.6%)	<.001

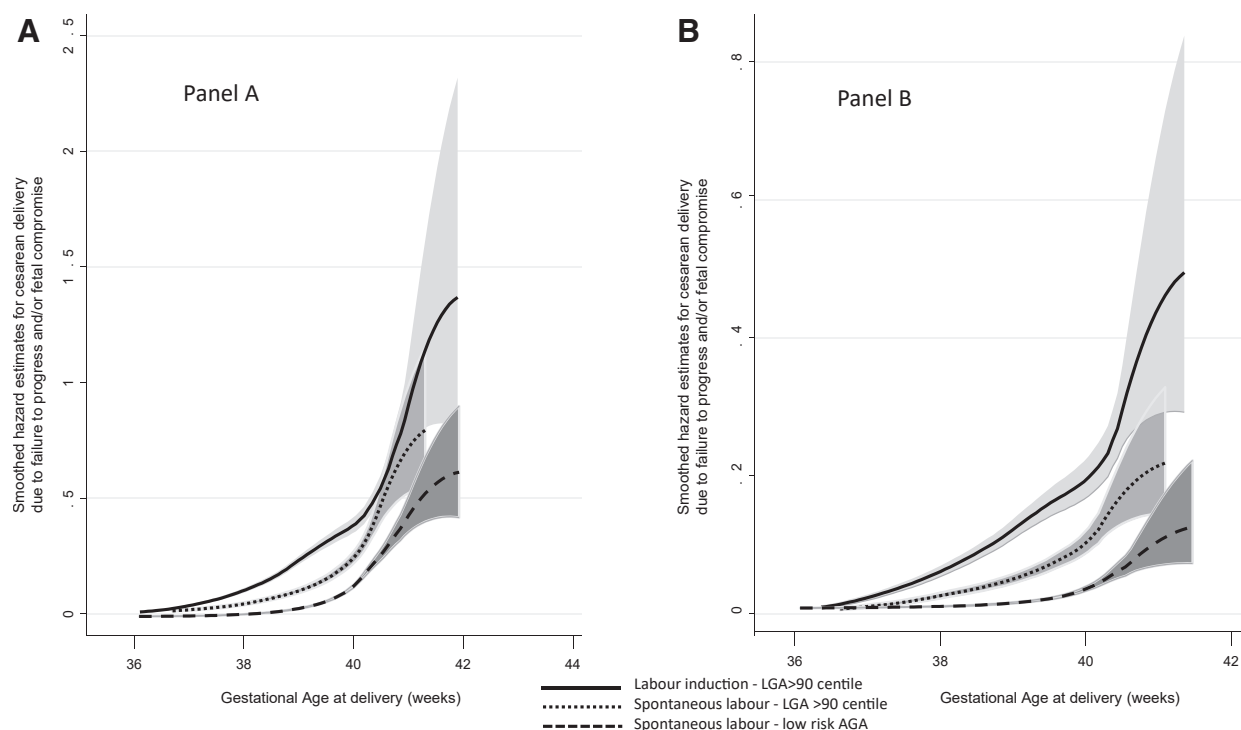
Other patterns exist with lower frequencies. The percentages describe the proportion of women within each risk-factor pattern who experienced spontaneous vs induced labor.

BMI, body mass index; CH, chronic hypertension; DM, diabetes mellitus; EFW, estimated fetal weight; GDM, gestational diabetes mellitus; GH, gestational hypertension; PE, preeclampsia.

^a χ^2 test.

FIGURE 1

Smoothed hazard of cesarean for failure to progress/fetal compromise



Smoothed instantaneous hazard estimate of cesarean delivery due to failure to progress and/or fetal compromise in nulliparous (panel A) and parous women (panel B). Shaded areas represent 95% confidence intervals; estimates are shown for descriptive purposes. Wider bands for induction and at later gestational ages reflect clinical heterogeneity and fewer observations.

category, when compared specifically to spontaneous labor occurring at or beyond 42⁺⁰ weeks.

Induction of labor beyond 41 weeks

IOL beyond 41 weeks is not recommended in LGA, as it was consistently associated with a higher incidence of cesarean delivery for fetal compromise and/or failure to progress compared to spontaneous labor at any GA (Figures 2 and 3, A and B).

The relative benefit of planned IOL in reducing cesarean delivery for fetal compromise and/or failure to progress was greater among nulliparous compared to parous, across both EFW categories.

The output of the competing risk regression analysis is shown in Supplemental Table 1, providing insights into the timing of IOL in relation to the CIF of cesarean delivery for women with LGA fetuses. As shown, using the whole population of fetuses with EFW between the 10th and 90th percentile as the

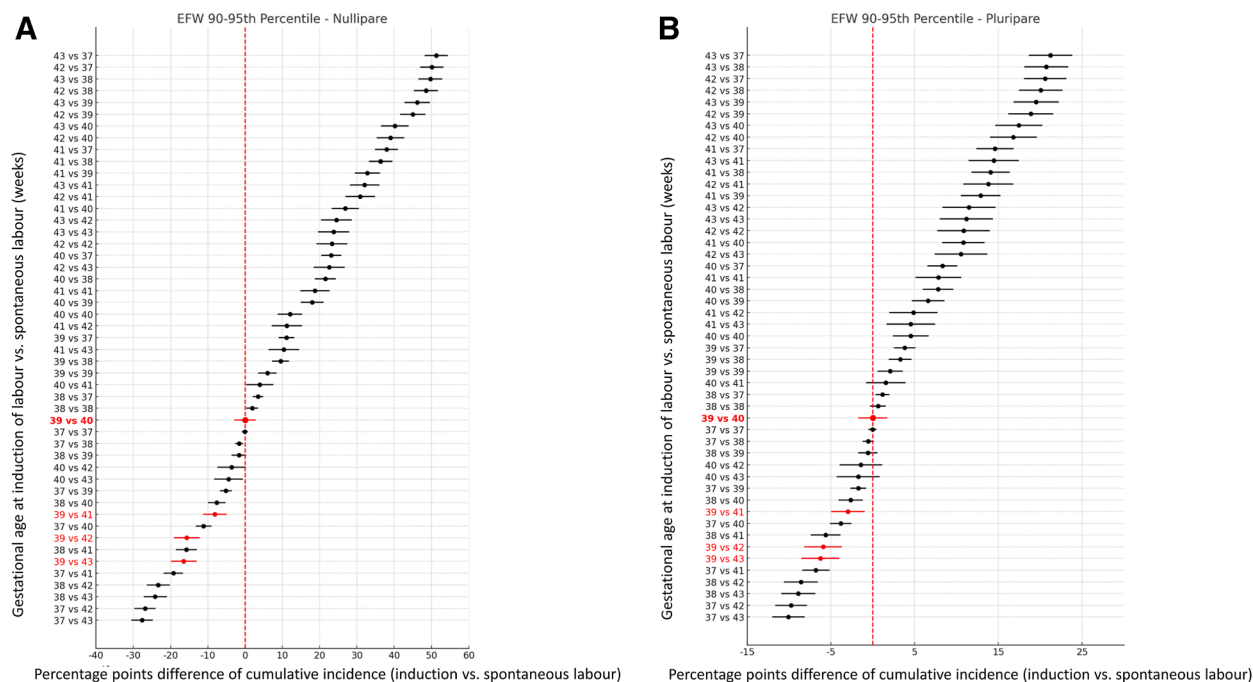
reference group, increasing EFW was associated with a progressively higher subhazard ratio (SHR) for cesarean delivery due to fetal compromise and/or failure to progress. Specifically, the SHR was 1.62 for fetuses with EFW between the 90th and 95th percentile and 2.05 for those above the 95th percentile, indicating a 62% and 105% increase, respectively, in the CIF. This effect is consistent in both nulliparous and parous. Parity and mode of labor initiation were identified as important risk factors, with lower SHR values observed in cases of spontaneous labor (SHR, 0.66) and higher values for nulliparous women (SHR, 3.05). Parity emerged as the most important risk factor for cesarean delivery due to fetal compromise and/or failure to progress. Additionally, lower maternal height (particularly when <162 cm) was found to impact the SHR for cesarean delivery risk. This confirmed a graded inverse association between increasing height and lower cesarean

delivery risk. Furthermore, in parous, the interpregnancy interval (in years) emerged as an independent risk factor for cesarean delivery (SHR, 1.07; 95% confidence interval, 1.050–1.095; $P < .001$; data not shown).

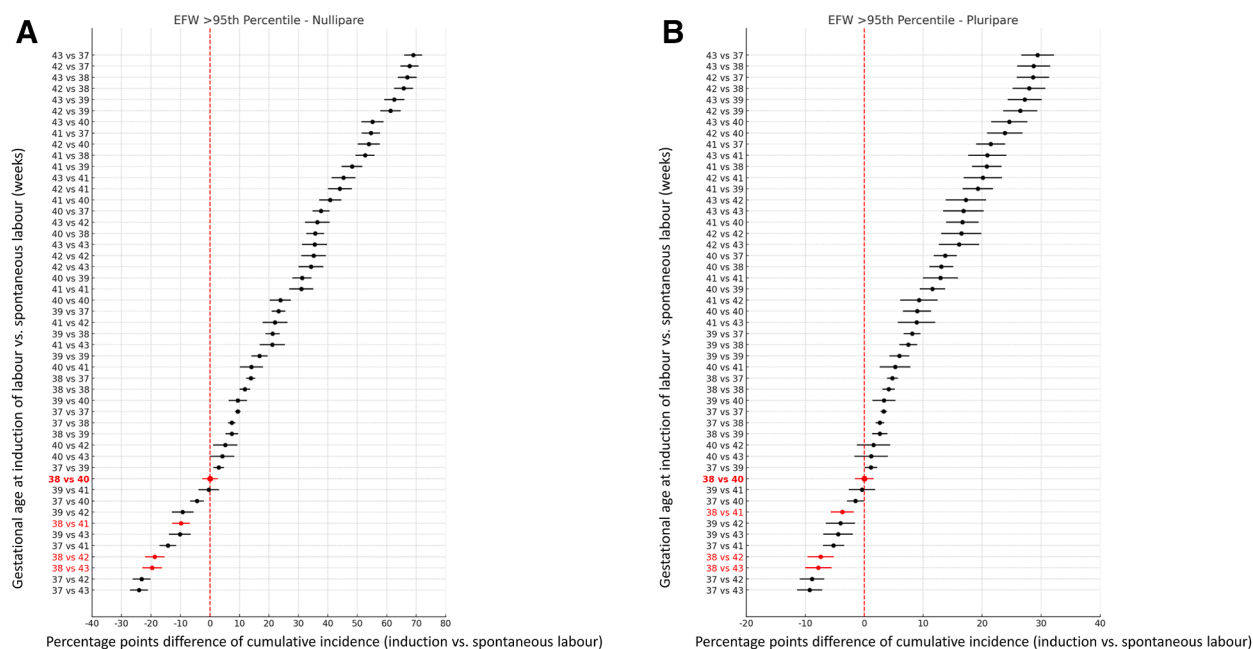
Supplemental Table 2 reports the concordance between EFW >90th centile and neonatal BW >90th centile, stratified by the interval between the EFW assessment and delivery, and by neonatal BW category. The highest concordance (approximately 80%) was observed among infants with BW >95th centile delivered within 10 days of the EFW assessment. Concordance decreased for deliveries occurring more than 10 days after the assessment and for neonates with BW in the 90th to 95th centile range.

Comment Principal findings

This study showed that: first, the hazard of cesarean delivery due to fetal

FIGURE 2**Cesarean incidence by gestational age and labor onset (LGA 90–95th)**

Percentage points differences (95% CI) in cumulative incidence of cesarean delivery between induction of labor and spontaneous labor at various gestational ages, for fetuses with estimated fetal weight percentile from 90th to the 95th percentiles in nulliparous (panel **A**) and parous women (panel **B**). Negative differences favor induction; positive differences favor spontaneous labor.

FIGURE 3**Cesarean incidence by gestational age and labor onset (LGA >95th)**

Percentage points differences (95% CI) in cumulative incidence of cesarean delivery between induction of labor and spontaneous labor at various gestational ages, for fetuses with estimated fetal weight percentile above the 95th percentile in nulliparous (panel **A**) and parous women (panel **B**). Negative differences favor induction; positive differences favor spontaneous labor.

compromise and/or failure to progress increases progressively with advancing GA with steeper rise after 40 weeks; second, at a given GA, the hazard is higher in nulliparous vs parous women and in IOL vs spontaneous labor; third, there is a gradient of hazard progression when EFW increases from the 90th to above the 95th percentile; fourth, a policy of IOL at 38 to 39 weeks for fetuses with EFW >95th or 90th to 95th percentile approximates the hazard and cumulative incidence of LGA fetuses delivering spontaneously at 40 weeks; and fifth, induction beyond 41 weeks is associated with a substantially increased risk.

The analysis compares the risk of cesarean delivery due to fetal compromise and/or failure to progress observed in case IOL at a given GA with ongoing expectant management, accounting for spontaneous labor as an intrinsic component of the expectant management pathway. Nulliparous, who are generally at a higher baseline risk for intrapartum complications, may derive more substantial gains from a timely IOL strategy, as compared to parous.

The findings support IOL at 38 weeks in cases of EFW >95th percentile and at 39 weeks for EFW between the 90th and 95th percentile. Alternatively, conservative management until 41 weeks may be considered, followed by elective cesarean delivery if spontaneous labor has not occurred by then, to avoid potentially harmful effects of late induction.

Results in the context of what is known

Routine ultrasonographic estimation of EFW at 36 weeks' gestation is effective in the prediction of LGA neonates^{19–21} and the risk of such pregnancies is well recognized.^{1–7} However, management remains controversial and the role of routine IOL in LGA remains debated, partly depending on the specific outcome to be prevented. Some high-quality studies support early delivery,^{5,6,22,23} while major guidelines generally do not^{14,24,25} despite increasing safety, broader adoption, and growing evidence that elective IOL at 39 weeks' gestation improves maternal and

neonatal outcomes even in low-risk pregnancies.^{26,27}

A recent major trial from the United Kingdom showed that IOL in LGA fetuses with EFW >90th percentile associates with risk reduction for shoulder dystocia and cesarean delivery while ensuring a shorter second stage of labor.⁶ Prolonged labor duration (particularly the second stage) is associated with a greater risk of neonatal acidemia regardless of cardiotocographic findings.²⁸ Moreover, given that that IOL in the low-risk population shows progressively safer outcomes,^{26,27} it is reasonable to hypothesize a role of IOL in fetuses with LGA, as well. In fact, different studies assessed the role of IOL in LGA after assessment by ultrasound^{5,6,22,23} and magnetic resonance,²⁹ showing improved perinatal outcomes. A detailed discussion on the method of IOL is beyond the scope of this study; however, it is certainly relevant given the evidence of lower risk of adverse perinatal events with mechanical methods as compared to prostaglandins.³⁰

Strengths and limitations

Strengths include the well-defined sizable study cohort, supporting the generalizability of the findings. Additionally, the involvement of highly trained operators under standardized protocols, combined with formal quality control at a leading obstetric research center (FMF), ensured high-quality data collection. Importantly, listwise deletion for missing data affected less than 1% of cases, minimizing the risk of selection bias.

Limitations include the retrospective design and lack of additional maternal outcomes (eg, postpartum hemorrhage, patient satisfaction) or neonatal complications (eg, low Apgar scores, acidemia). Despite some data suggesting that IOL does not increase these risks, future intervention studies may assess these outcomes. Since the database includes only pregnancies undergoing routine third-trimester ultrasound assessment at 35⁺⁰ to 36⁺⁶ weeks' gestation, the proportion of women delivering beyond this window without

EFW assessment cannot be reliably estimated. A substantial proportion of prelabor cesarean deliveries involved suspected LGA fetuses based on antenatal ultrasound findings, suggesting that knowledge of EFW may have influenced clinical management. Any nondifferential misclassification between EFW and BW categories would be expected to attenuate observed associations. Finally, induction indications may reflect populations with different baseline risks, partly explaining the hazard plateau at 40 weeks and sharp rise after 41 weeks.

Clinical implications and future research

To equalize the risk of cesarean delivery due to failure to progress and/or fetal compromise in LGA after IOL with that of spontaneous labor at term, there are 2 options to be considered as comparative group: either the low-risk AGA population or the LGA group itself. The GA at IOL in the LGA group will naturally be lower when using AGA vs LGA as a comparator group, creating a GA range for IOL that typically ranges from 38 to 39 weeks. The timing of IOL in LGA pregnancies should be determined by both the EFW percentile and the level of cesarean delivery risk deemed acceptable, with higher EFW and lower expected risk favoring earlier induction, around 38 weeks. Although this analysis focuses on intrapartum cesarean delivery, the GAs at which induction appears most favorable are within ranges deemed safe for the fetus in large, randomized trials; nonetheless, fetal safety outcomes were not directly assessed.

Previous research evidence showed a strong association between placental function represented by the FMF risk of PE or by the ratio between soluble fms-like tyrosine kinase-1 and placental growth factor and intrapartum fetal compromise.^{16–18} Therefore, future prediction models may reassess this knowledge and provide a comprehensive method encompassing all variables.

Conclusion

This study provides insights into the optimal birth timing of LGA suggesting

that IOL at 39 weeks (EFW 90th–95th percentile) or 38 weeks (EFW >95th percentile) can substantially reduce the incidence of cesarean delivery due to fetal compromise and/or failure to progress, aligning it with that observed in spontaneous labor at 40 weeks. The relative benefit of planned IOL was greater among nulliparous compared to parous, particularly with EFW >95th percentile, reflecting their higher baseline risk of intrapartum cesarean delivery and steeper increase in risk with advancing GA. Nevertheless, the overall cesarean risk remains elevated in IOL, especially for the largest fetuses, emphasizing the importance of careful selection and timing.

This is relevant given that LGA fetuses face a progressively higher risk of cesarean delivery beyond 40 weeks, irrespective of parity or labor onset. The evidence from this study advises against IOL in LGA pregnancies beyond 40 weeks, as it offers no benefit and entails substantial risks after 41 weeks. This study highlights the need to tailor IOL timing to fetal size, GA, and maternal factors, especially before 40 weeks, to balance maternal and neonatal risks and maintain cesarean rates comparable to spontaneous labor. Findings should be interpreted as hypothesis-generating rather than prescriptive. Future research should refine these recommendations and develop more individualized strategies for LGA pregnancies. ■

References

- Hocquette A, Durox M, Wood R, et al. International versus national growth charts for identifying small and large-for-gestational age newborns: a population-based study in 15 European Countries. *Lancet Reg Health Eur* 2021;8:100167.
- Koyanagi A, Zhang J, Dagvadorj A, et al. Macrosomia in 23 developing countries: an analysis of a multicountry, facility-based, cross-sectional survey. *Lancet* 2013;381:476–83.
- Beta J, Khan N, Khalil A, Fiolna M, Ramadan G, Akolekar R. Maternal and neonatal complications of fetal macrosomia: systematic review and meta-analysis. *Ultrasound Obstet Gynecol* 2019;54:308–18.
- Beta J, Khan N, Fiolna M, Khalil A, Ramadan G, Akolekar R. Maternal and neonatal complications of fetal macrosomia: cohort study. *Ultrasound Obstet Gynecol* 2019;54:319–25.
- Boulvain M, Senat MV, Perrotin F, et al. Induction of labour versus expectant management for large-for-date fetuses: a randomised controlled trial. *Lancet* 2015;385:2600–5.
- Gardosi J, Ewington LJ, Booth K, et al. Induction of labour versus standard care to prevent shoulder dystocia in fetuses suspected to be large for gestational age in the UK (the big baby trial): a multicentre, open-label, randomised controlled trial. *Lancet* 2025;405:1743–56.
- Robertson K, Vieira M, Impey L. Perinatal outcome of fetuses predicted to be large-for-gestational age on universal third-trimester ultrasound in non-diabetic pregnancy. *Ultrasound Obstet Gynecol* 2024;63:98–104.
- Hadlock FP, Harrist RB, Sharman RS, Deter RL, Park SK. Estimation of fetal weight with the use of head, body, and femur measurements—a prospective study. *Am J Obstet Gynecol* 1985;151:333–7.
- von Elm E, Altman DG, Egger M, et al. The Strengthening of reporting of observational studies in epidemiology (STROBE) statement: guidelines for reporting observational studies. *Lancet* 2007;370:1453–7.
- Robinson HP, Fleming JE. A critical evaluation of sonar "crown-rump length" measurements. *Br J Obstet Gynaecol* 1975;82:702–10.
- Nicolaides KH, Wright D, Syngelaki A, Wright A, Akolekar R. Fetal medicine foundation fetal and neonatal population weight charts. *Ultrasound Obstet Gynecol* 2018;52:44–51.
- Hamilton EF, Zhoroev T, Warrick PA, et al. New labor curves of dilation and station to improve the accuracy of predicting labor progress. *Am J Obstet Gynecol* 2024;231:1–18.
- American College of Obstetricians and Gynecologists (College); Society for Maternal-Fetal Medicine, Caughey AB, Cahill AG, Guise JM, Rouse DJ. Safe prevention of the primary cesarean delivery. *Am J Obstet Gynecol* 2014;210:179–93.
- National Institute for Health and Care Excellence. In: Inducing labour. NICE guideline [NG207]. London: NICE. 2021. Available from: <https://www.nice.org.uk/guidance/ng207>.
- Romero R, Sabo Romero V, Kalache KD, Stone J. Parturition at term: induction, second and third stages of labor, and optimal management of life-threatening complications—hemorrhage, infection, and uterine rupture. *Am J Obstet Gynecol* 2024;230:S653–61.
- Farina A, Cavoretto PI, Syngelaki A, Morano D, Adjahou S, Nicolaides KH. The 36-week preeclampsia risk by the fetal medicine foundation algorithm is associated with fetal compromise following induction of labor. *Am J Obstet Gynecol* 2024. <https://doi.org/10.1016/j.ajog.2024.12.025>.
- Farina A, Cavoretto PI, Syngelaki A, Adjahou S, Nicolaides KH. Soluble fms-like tyrosine kinase-1/placental growth factor ratio at 36 weeks' gestation: association with spontaneous onset of labor and intrapartum fetal compromise in low-risk pregnancies. *Am J Obstet Gynecol* 2025;232:392.e1.
- Cavoretto PI, Farina A, Syngelaki A, Solonos F, Nicolaides KH. The risk for pre-eclampsia at 36 weeks and at 12 weeks predicts intrapartum fetal compromise and spontaneous onset of labor. *Am J Obstet Gynecol* 2025. <https://doi.org/10.1016/j.ajog.2025.04.042>.
- Khan N, Ciobanu A, Karampitsakos T, Akolekar R, Nicolaides KH. Prediction of large-for-gestational-age neonate by routine third-trimester ultrasound. *Ultrasound Obstet Gynecol* 2019;54:326–33.
- Ewington LJ, Hugh O, Butler E, Quenby S, Gardosi J. Accuracy of antenatal ultrasound in predicting large-for-gestational-age babies: population-based cohort study. *Am J Obstet Gynecol* 2025;232:210.e1.
- Frick AP, Syngelaki A, Zheng M, Poon LC, Nicolaides KH. Prediction of large-for-gestational-age neonates: screening by maternal factors and biomarkers in the three trimesters of pregnancy. *Ultrasound Obstet Gynecol* 2016;47:332–9.
- Boulvain M, Thornton JG. Induction of labour at or near term for suspected fetal macrosomia. *Cochrane Database Syst Rev* 2023;3:CD000938.
- Badr DA, Carlin A, Kadji C, Kang X, Cannie MM, Jani JC. Timing of induction of labor in suspected macrosomia: retrospective cohort study, systematic review and meta-analysis. *Ultrasound Obstet Gynecol* 2024;64:443–52.
- Macrosomia: ACOG practice bulletin, number 216. *Obstet Gynecol* 2020;135:e18–35.
- The royal Australian and New Zealand College of Obstetricians and Gynaecologists (RANZCOG). Diagnosis and management of suspected fetal macrosomia. 2021. <https://ranzco.org.au/wp-content/uploads/Diagnosis-Management-Suspected-Fetal-Macrosomia.pdf>. Accessed May 5, 2025.
- Sotiriadis A, Petousis S, Thilaganathan B, et al. Maternal and perinatal outcomes after elective induction of labor at 39 weeks in uncomplicated singleton pregnancy: a meta-analysis. *Ultrasound Obstet Gynecol* 2019;53:26–35.
- Grobman WA. The role of labor induction in modern obstetrics. *Am J Obstet Gynecol* 2024;230:S662–8.
- Cavoretto PI, Seidenari A, Amodeo S, et al. Quantification of posterior risk related to intrapartum FIGO 2015 criteria for cardiotocography in the second stage of labor. *Fetal Diagn Ther* 2021;48:149–57.
- Badr DA, Cannie MM, Kadji C, Kang X, Carlin A, Jani JC. Reducing macrosomia-related birth complications in primigravida women: ultrasound- and magnetic resonance imaging-based models. *Am J Obstet Gynecol* 2024;230:557.e1.
- Jones MN, Palmer KR, Pathirana MM, et al. Balloon catheters versus vaginal prostaglandins

for labour induction (CPI collaborative): an individual participant data meta-analysis of randomised controlled trials. *Lancet* 2022;400:1681–92.

Author and article information

From the Obstetric Unit, Istituto di Ricovero e Cura a Carattere Scientifico Azienda Ospedaliero-Universitaria di Bologna, Bologna, Italy (Farina); Department of Medical and Surgical Sciences, Alma Mater Studiorum, University of Bologna, Bologna, Italy (Farina); Department of Ob-

stetrics and Gynaecology, IRCCS San Raffaele Scientific Institute, Milan, Italy (Cavoretto); School of Medicine, Vita-Salute San Raffaele University, Milan, Italy (Cavoretto); Fetal Medicine Research Institute, King's College Hospital, London, UK (Syngelaki, Mitrogiannis and Nicolaidis); Fetal Medicine Unit, Medway Maritime Hospital, Gillingham, UK (Akolekar); and Institute of Medical Sciences, Canterbury Christ Church University, Chatham, UK (Akolekar).

Received Nov. 4, 2025; revised March 8, 2026; accepted March 24, 2026.

The authors report no conflict of interest.

The study was supported by a grant from the Fetal Medicine Foundation (UK Charity No.: 1037116).

This study was conducted in accordance with the ethical standards for human research established by the Declaration of Helsinki. The original data collection received approval from the London—Dulwich Research Ethics Committee (reference number is 02-03-033).

Corresponding author: Paolo Ivo Cavoretto, MD, PhD. cavoretto.paolo@hsr.it